INCH-POUND

MIL-PRF-1/1295E
19 July 1999
SUPERSEDING
MIL-E-1/1295D
30 March 1983

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING
TYPE 6842

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Pentode, high voltage, sharp cutoff.

Outline --- 5-2 (EIA) (see figure 1 for dimensions).

Base --- E7-1.

Envelope --- T5-1/2.

Cathode --- Coated unipotential.

Base connections:

Pin no.	1	2	3	4	5	6	7	Cap
Element	G2	H	K, G3	G1	K, G3	H	K, G3	a

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rg1	Ib(ave)	ib	Pp	Pg2
Unit:	V	V dc	V dc	V dc	V dc	Meg	ma	ma	W	W
Maximum:	6.9	4,000	---	150	±300	0.1	10	100	8	.5
Minimum:	5.7	---	-100	---	---	---	---	---	---	---
Test conditions:	6.3	1,500	-1.5	100	---	---	---	---	---	----

See footnotes at end of table I.

GENERAL: Qualification - Not required.

First article testing: Required (See MIL-PRF-1, appendix F).

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TABLE I. Testing and inspection.

Inspection	Method	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1								
Heater current	1301		---	---	If	135	165	mA
Heater-cathode leakage	1336	Ehk = ±100	1	I	Ihk	---	50	µA dc
Total grid current	1266	1/	1	I	Ic1	---	-0.3	µA dc
Electrode current (1) (anode)	1256	1/	1	I	Ib	2.6	4.5	mA dc
Electrode current (screen grid)	1256		1	I	Ic2	---	1.0	mA dc
Transconductance	1306		1	I	Sm	2,050	3,650	µmhos
Conformance inspection, part 2								
Electrode current (2) (anode) (idling)	1256	Ec1 = -10 V dc	2.5	---	Ibo	---	100	µA dc
Direct-interelectrode capacitance	1331	Unshielded	---	---	Cgp Cin Cout	--- 3.0 17.0	.040 5.0 2.3	PF pF pF
High-frequency vibration	1031	Rp = 2,000 ohms; no voltages	---	2/	Ep	---	200	mV ac
Insulation of electrodes	1211		---	2/	---	---	---	---
Permanence of marking	1105		---	---	---	---	---	---
Shock	1041	30° hammer angle	---	2/	---	---	---	---
Conformance inspection, part 3								
Intermittent life	1501	Group D Eb = 3,000 V dc; Ec2 = 100 V dc; RK = 2,500 ohms; Ehk = +100 V dc	---	---	t	500	---	Hrs
Intermittent life test end points (500 hours)	---						-	
Electrode current (1) (anode)	1256		---	---	---	2.0	---	mA dc
Transconductance (change of individual tubes)	1306		---	---	ΔSm t	---	25%	µmhos

1/ This test shall be performed at the conclusion of the holding period.
2/ This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
3/ Revision letters are not used to denote changes due to the extensiveness of the changes.

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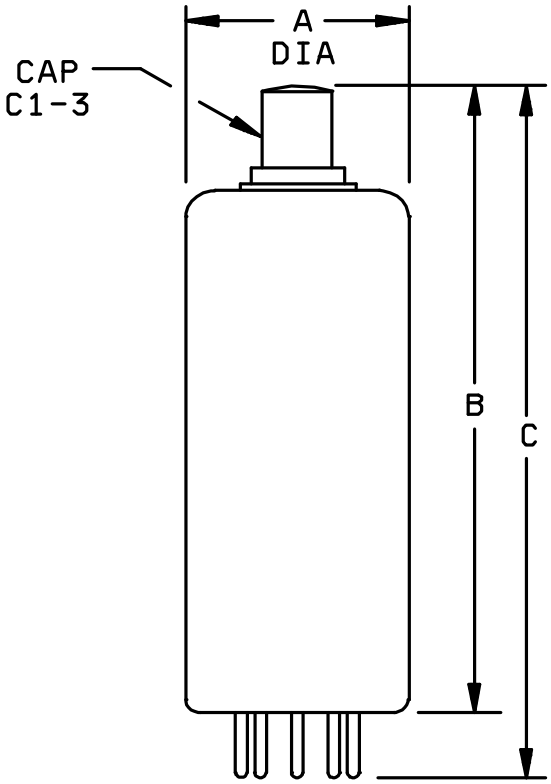

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	---	.750 DIA	---	19.05 DIA
B	---	2.000	---	50.80
C	---	2.250	---	57.15

FIGURE 1. Outline drawing for type 6842.

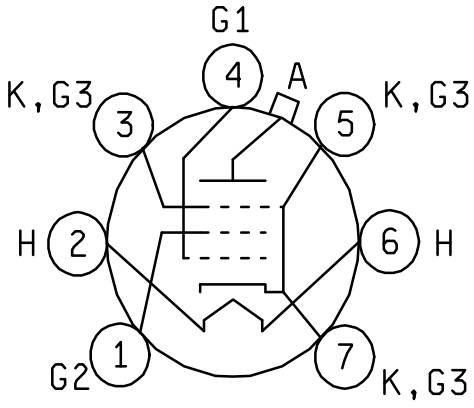

FIGURE 2. Schematic for type 6842.

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Air Force - 11
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Review activities:
Air Force - 99

Preparing activity:
DLA - CC

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